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INCH-POUND

MIL-STD-001303C(OS)
11 APRIL 1994
USED IN LIEU OF
MIL-STD-1303B
15 APRIL 1975

DEPARTMENT OF DEFENSE
STANDARD PRACTICE

PAINTING OF
NAVAL ORDNANCE EQUIPMENT

AMSC N/A

AREA MISC

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FORWARD

1. This military standard has been prepared by the Naval Sea Systems Command based upon currently available technical information but it has not been approved for promulgation as a coordinated revision of MIL-STD-1303. It is subject to modification. However, pending its promulgation as a coordinated military standard, it may be used.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Standardization Branch (Code 8420), 101 Strauss Avenue, Indian Head, MD 20640-5035 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or sending a letter.

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1. SCOPE

1.1 Scope. This standard covers materials, methods, and processes for paint systems approved for use in the painting and protection of various types of naval ordnance equipment and weapons against deterioration (table I). The term 'painting' is understood to include such preparatory treatments as cleaning, rust removal, surface treatments, filling, caulking, and the subsequent application of coating such as pretreatments, primers, final coats of paint, enamel, varnish, and other organic protective films. The paint systems described herein do not include color-coding or other data to identify the characteristics of weapons, equipment, components, or fillers. (References should be made to applicable standards, specifications, or drawings for color-codes to be used on specific items.)

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

TT-C-490	Cleaning Methods for Ferrous Surfaces and Pretreatments for Organic Coatings
TT-F-336	Filler, Wood, Paste
TT-P-28	Paint, Aluminum, Heat Resisting (1200°F)
TT-P-595	Preservative Coating, Canvas
TT-P-645	Primer, Paint, Zinc-Molybdate, Alkyd Type
TT-P-662	Primer Surfacer, Sanding, Lacquer and Enamel Type
TT-V-119	Varnish, Spar, Phenolic-Resin
TT-W-571	Wood Preservation: Treating Practices
TT-W-572	Wood Preservative: Water Repellent

MILITARY

MIL-V-173	Varnish, Moisture and Fungus Resistant (For Treatment of Communications, Electronics and Associated Equipment)
MIL-C-450	Coating Compound, Bituminous Solvent Type, Black (For Ammunition)
MIL-R-3043	Resin Coating, Permanent, For Engine Components and Metal Parts
MIL-C-3301	Compound, Asphaltic, Hot Melt (Cavity Lining)
MIL-L-3891	Luminescent Material and Equipment (Nonradioactive)

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MIL-C-4556	Coating Kit, Epoxy, For Interior of Steel Fuel Tanks
MIL-S-5002	Surface Treatments and Inorganic Coatings for Metal Surfaces of Weapons Systems
MIL-F-7179	Finishes, Coatings, and Sealants for the Protection of Aerospace Weapons Systems
MIL-S-8802	Sealing Compound, Temperature-Resistant, Integral Fuel Tanks and Fuel Cell Cavities, High Adhesion
MIL-C-10578	Corrosion Removing and Metal Conditioning Compound (Phosphoric Acid Base)
MIL-P-15931	Paint, Antifouling, Vinyl (Formulas No. 121, 121A, 129, and 129A)
MIL-C-16173	Corrosion Preventive Compound, Solvent Cutback, Cold-Application
MIL-C-18255	Caulking Compound, Synthetic Rubber Base, Wooden Deck Seam Application
MIL-F-18264	Finishes: Organic, Weapons System, Application and Control of
MIL-C-18487	Compound, Gun Slushing
MIL-H-18911	Heads and Motors, Rocket, Inert Parts, Painting and Marking
MIL-P-18948	Projectiles, Painting, Marking, and Greasing of
MIL-P-21960	Painting and Marking; Warheads, Guided Missile
MIL-C-22750	Coating, Epoxy, VOC - Compliant
MIL-T-23142	Tape, Pressure-Sensitive Adhesive for Dissimilar Metal Separation
MIL-G-23827	Grease, Aircraft and Instrument, Gear and Actuator Screw, NATO Code Number G-354, Metric
MIL-I-24092	Insulating Varnishes and Solventless Resins for Application by the Dip Process
DOD-C-24176	Cement, Epoxy, Metal Repair and Hull Smoothing
MIL-I-24391	Insulation Tape, Electrical, Plastic, Pressure-Sensitive

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MIL-P-24441	Paint, Epoxy-Polyamide, General Specification For
MIL-P-24441/1	Paint, Epoxy-Polyamide, Green Primer, Formula 150, Type I
MIL-P-24441/6	Paint, Epoxy-Polyamide, Exterior Topcoat, Dark Gray, Formula 155-RO = 6, Type I
DOD-E-24607	Enamel, Interior, Nonflaming (Dry), Chlorinated Alkyd Resin, Semigloss (Metric)
MIL-E-24635	Enamel, Silicone Alkyd Copolymer
MIL-P-24647	Paint System, Anticorrosive and Antifouling, Ship Hull
DOD-P-24648	Primer Coating, Zinc Dust Pigmented for Exterior Steel Surfaces
MIL-P-53030	Primer Coating, Epoxy, Water Reducible, Lead and Chromate Free
MIL-G-81322	Grease, Aircraft, General Purpose, Wide Temperature Range
MIL-C-85285	Coating: Polyurethane, High-Solids

STANDARDS

FEDERAL

FED-STD-141	Paint, Varnish, Lacquer and Related Materials: Methods of Inspection, Sampling and Testing
FED-STD-595	Colors Used in Government Procurement

MILITARY

MS603	Formula for Fuze Thread Luting (For Special Ammunition Use)
MIL-STD-709	Ammunition Color Coding
MIL-STD-889	Dissimilar Metals

(Unless otherwise indicated, copies of Federal and military specifications, and standards are available from: Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the even of a conflict between the text of this document and the

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references cited herein, the text of this document takes precedence. Nothing in this document; however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. DEFINITIONS

3.1 Paint systems. Paint systems are the series of coatings applied over properly cleaned and chemically treated or otherwise pretreated surfaces. A paint system will include the following, in the order of application:

a. Type of primer and number of coats each, in the order of application to the surfaces including any fillers, surfacers, or sealers required.

b. Type of finish coats (paints, enamels, varnishes, or others) and number of coats each, in the order of application over the primer.

c. Thickness of individual coats or of the entire system and other pertinent information is included where necessary.

3.2 Resins. Resins are solid organic substances, either natural, originating in the secretion of certain plants and insects, or synthetic. They are thermoplastic, flammable, nonconductive of electricity, break with a conchoidal fracture (when hard), and dissolve in certain specific organic solvents other than water.

3.3 Drying oils. Drying oils, as used in paint compositions, possess (to a marked degree) the property of readily taking up oxygen from the air and changing to a relatively hard, tough elastic substance when exposed in a thin film in air.

3.4 Pigments. Pigments are fine solid particles that impart hiding and color to paint and are substantially insoluble in the vehicle.

3.5 Primers. Primers are the first coat of a paint system. They are applied to suitably prepared wood, or to chemically or electrochemically treated metal. They improve adhesion of the subsequent paint coats and aid in protecting the metal against corrosion.

3.6 Fillers. Fillers are pigmented compositions for filling pores or irregularities in a surface preparatory to application of other finishes.

3.7 Surfacers. Surfacers are pigmented compositions for filling minor irregularities to obtain a smooth uniform surface preparatory to applying finish coats. They are usually applied over a primer and sanded for smoothness.

3.8 Sealers. Sealers are liquid compositions that prevent excess absorption of finish coats into porous surfaces and can also be used to prevent bleeding.

3.9 Paints. Paints are pigmented liquid compositions that are converted to an opaque solid film after application as a thin layer.

3.10 Enamels. Enamels are paints characterized by their ability to form an especially smooth film.

3.11 Varnishes. Varnishes are solutions of resins and drying oils, or resins alone in solvents. Upon

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drying, they form a clear to dark amber film. Varnishes may be mixed with stains to approximately duplicate wood finishes. Varnishes that contain a large amount of opaque pigments are usually listed as enamels.

3.12 Other terms. Other general terms, frequently encountered in the specifications of paints, varnishes, and related materials may be found in Section 8 of FED-STD-141.

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4. GENERAL REQUIREMENTS

4.1 Use of this standard. This standard shall be used as a general guide to the selection of painting materials, systems, procedures, and their designation or identification on drawings or other documents applicable to naval ordnance equipment and weapons. For general procedures, such as preparation for painting, reference may be made to the standard as a whole. However, specific reference shall be made, in accordance with the descriptions or designations herein, when particular materials, processes, or paint systems are to be used. System number and alternates, if any, must be specified when this standard is used to define a protective finish on a hardware item.

4.2 Use of specification materials and processes. Unless otherwise specified in the procurement documents, materials, and processes used in painting of naval ordnance equipment shall be in accordance with the latest issues of the applicable specifications listed herein. The film thickness of paint type materials and techniques of the application shall be as specified in MIL-F-18264. Other Federal and military specifications may be used when prior approval is obtained from the Naval Sea Systems Command. Company standards, industry specifications and industry practices shall be subject to prior approval by the Naval Sea Systems Command before use.

4.3 Surfaces requiring painting. Practically all metal and wood surfaces exposed to weather are painted for protection against deterioration. Surfaces of some metals, such as corrosion-resisting steel, copper alloys, and certain aluminum containers may be left unpainted if they are enclosed or otherwise sufficiently protected against moisture, high humidity, or salt spray. Aluminum containers, where determined necessary, may be coated with primer conforming to MIL-P-24441.

4.4 Surfaces not to be painted. Certain types of surfaces do not require painting or are harmfully affected by painting. These include the following:

a. Machined surfaces that are required to move against or assemble to others such as threads, slides, bearing contacts, gear teeth, bourrelets, rotating bands, and linkages. Moving parts are generally lubricated when in use.

b. Electrical parts and components already assembled, such as contact supports, relays, sockets, plugs, connectors, and insulators. When any portion of these are to be painted, it shall be done prior to assembly. If touchup is required, as on cut or machined plastics, utmost care shall be used to avoid leaving any paint residue on electrical contact surfaces, hinges, pivots, bearing surfaces, or others whose proper operation may be affected by such residues.

c. Parts of copper or copper alloys, plated or unplated, such as electrical sockets, plugs, connectors, contacts, and terminals.

d. Plastic, rubber, or ceramic insulators, mounts, spacers, gaskets, or windshields.

e. Lubrication fittings, oil cups, and grease fittings. Adjacent piping may be painted or color-coded.

f. Identification, instruction, and warning plates.

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4.5 Preservation of unpainted surfaces.

4.5.1 Permanent assemblies. When required by the applicable drawings or specifications, mating threaded, bearing, and seating surfaces of metallic parts of weapons that are to be permanently assembled shall be coated before assembled with luting conforming to MS603. After assembly, the cleanliness of adjacent surfaces to be painted shall conform to TT-C-490. See 5.6.2.1 for treatment of other faying surfaces.

4.5.2 Subject to disassembly. Mating threaded, bearing, and seating surfaces of metallic parts that are not immediately assembled, or require disassembly during loading with explosives or other fillers, shall be coated before assembly with preservative grease conforming to MIL-G-81322. A thin continuous film shall be applied, avoiding contact of grease with any adjacent cavity surfaces to be painted.

4.5.3 Other unpainted surfaces. Metallic surfaces such as those of electrical contacts, bourrelets, retaining grooves, bearings, and gear teeth shall be coated with a thin, even, continuous film of corrosion-preventive compound conforming to grade 2 of MIL-C-16173. Interiors of gun barrels in storage shall be coated with gun slushing compound in accordance with MIL-C-18487 and protected against corrosion due to the entrance of water.

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5. DETAILED REQUIREMENTS

5.1 Preparation for painting.

5.1.1 Metallic surfaces. Before application of any pretreatment, primer, paint, or preservative, metallic surfaces shall be clean, dry, and free of oils, grease, dirt, finger prints, peeling paint, scale, rust, and any other foreign material. Surfaces of ferrous metals, aluminum and its alloys, magnesium and its alloys, titanium and its alloys, shall be cleaned and surfaces treated in accordance with MIL-F-7179 and the applicable requirements of MIL-S-5002. Copper and its alloys (brasses, bronzes, etc.) shall be treated in accordance with Type 1 of MIL-C-10578.

5.1.2 Nonmetallic Surfaces. Wood surfaces shall be clean, smooth, and sanded. Dents, holes, and checks shall be filled with wood filler conforming to TT-F-336. Wood surfaces shall be treated with a preservative in accordance with TT-W-571. Canvas shall be treated with a preservative in accordance with TT-P-595. Reinforced rubber and plastics are not usually treated, but cut surfaces may be coated with varnish conforming to MIL-V-173 in order to reduce moisture penetration.

5.2 Naval Sea Systems Command standard paint systems. Certain paint systems (see 3.1) have been established as standards for naval ordnance equipment. New systems are added as required, especially for missiles and other weapons. Certain systems have become obsolete or have been replaced by more up-to-date systems employing improved materials or providing greater resistance to deteriorating influences. Table I shows currently approved paint systems for naval ordnance equipment (see 6.3) and their principal uses. Table II, the index to table I, enables ready location of systems suitable for certain applications. Systems that have been discontinued, altered, or replaced by others since the previous issue are shown in table III.

TABLE I. *Naval Sea Systems Command standard paint systems.*

System No.	Name and Color	Primer, Filler, Surfacer[1]	Paint and Color No. FED-STD-595	No. of Coats	Uses
07	Enamel, dull black	TT-P-645	MIL-E-24635 Black, No. 37038	1	Interior nonbearing surfaces of optical instruments, for minimum reflectance
13	Enamel, lusterless and semigloss white	TT-P-645	MIL-E-24635 white, No. 37875 and white, No. 27875	1	Interior surfaces of mine cases (except where in contact with explosives) and instrument cases; exterior of missiles where specified.
14	Enamel, gloss white	TT-P-645	MIL-E-24635 white No. 17875	2	Metal surfaces not exposed to weather, for high light reflection

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TABLE I. *Naval Sea Systems Command standard paint systems* - Continued.

System No.	Name and Color	Primer, Filler, Surfacer[1]	Paint and Color No. FED-STD-595	No. of Coats	Uses
14C	Paint, Interior white, fire retardant	TT-P-645	DOD-E-24607	1	Metal interior surfaces not exposed to weather, for high light reflection and fire retarding
20	Coating system for topside combat systems, *equipment, haze* gray	MIL-P-24441/1 (1 coat, 3 to 4 mils) and then MIL-P-24441/6 (1 coat, 2 to 3 mils)	MIL-E-24635, color No. 26270 (2 coats, 1 to 1.5 mils each coat)	4 coats total, 7 to 10 mils total thickness	Exterior surfaces of above deck radar and optical equipment, excluding reflective surfaces of radomes, Mk 86 Remote Optical Sight, AN/SPQ-9, AN/SPG-60, AN/SPQ-9B
20A	Coating system for reflective surfaces, haze gray	MIL-P-24441/1 (1 coat)	Spraylat 599-Y1002 or equivalent (see 6.4) copper conductive paint (3 coats), MIL-P-24441/1 (1 coat), and then MIL-E-24635, color No. 26270 (2 coats)	8 coats total	Reflective surfaces, AN/SPG-60
20B	Coating system for radomes, haze gray	MIL-P-24441/1 (1 coat, 0.5 to 1 mil)	MIL-E-24635, color No. 26270 (2 coats, 1 to 1.5 mils each coat)	3 coats total, 2.5 to 4 mils total thickness	Exterior surface of radomes, AN/SPQ-9
22	Paint, haze gray (for metal)	TT-P-645	MIL-E-24635 color No. 26270	2	*Exterior of gun barrels,* mounts, launchers, depth charges, and projectors; cases for torpedo test equipment etc.
25	Paint, aluminum	TT-P-645	TT-P-28	2	Metal cases armored cable, ammunition and cartridge boxes and *tanks; exteriors of gun* barrels, mounts, launched depth charges, and projectors.
31	Varnish (for wood)	TT-F-336 filler	MIL-C-85285 Type II (Approved for use on wood surfaces and top coat only.), or TT-V-119	2	Natural wood finish, e.g., for hardwood boxes, and cases

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TABLE I. *Naval Sea Systems Command standard paint systems* - Continued.

System No.	Name and Color	Primer, Filler, Surfacer[1]	Paint and Color No. FED-STD-595	No. of Coats	Uses
37	Enamel, semigloss light gray	TT-P-645	MIL-E-24635 class II color, light gray No. 26307	2	Equipment inside shields, turrets or electrical panels and enclosures; cables, and enclosed fire control equipment
42	Varnish, electrical insulating	None	MIL-I-24092	1	Sealing, touchup, or moisture-proofing of electrical parts
[2]47	Coating compound, acidproof, cavity lining	None	MIL-C-450, solvent type, or MIL-C-3301 hotmelt	1	Surfaces of cavities in contact with explosives; acid resistant coating
48	Paint, anti-fouling red	MIL-P-24441	MIL-P-15931, or MIL-P-24647	2 or more to 4 mils minimum thickness	Exterior surfaces of moored mines, and floats; stencilling on moored mines (2)
48A	Enamel exterior white	MIL-P-24441	MIL-E-24635 color No. 27875	1 or more to 4 mils minimum thickness	Marking, lettering on mines
49	Paint, vinyl alkyd, gray	MIL-P-24441	MIL-E-24635 color No. 26118	to 4 mils minimum thickness	Floats, buoys, and harbor defense items (2)
51	Paint, fluorescent orange	TT-P-645	MIL-E-24635, MIL-L-3891 Type F, liquid form orange color. Topcoat MIL-C-85285, not more than 1 mil.	1 each (MIL-L-3891 not more than 3 mils)	Torpedo exercise head, for visibility; for underwater objects (distance sighting)
52	Enamel, semigloss green	TT-P-645 (1 coat)	MIL-E-24635 Type II green No. 24109 White No. 27875 Alternate green and white with final coat green	2 coats green 1 coat white	Exterior surfaces of aluminum torpedoes; metal surfaces of ammunition and ammunition containers
53	Paint vinyl alkyd gray	TT-P-645	MIL-E-24635 color No. 26118	1	Torpedoes
54	Coatings for color coding and marking	TT-P-645	MIL-E-24635, color as applicable	1	Color coding of weapons and ammunition components in accordance with MIL-STD-709, and lettering and stencil information

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TABLE I. *Naval Sea Systems Command standard paint systems* - Continued.

System No.	Name and Color	Primer, Filler, Surfacer[1]	Paint and Color No. FED-STD-595	No. of Coats	Uses
58	Enamel, quick drying lusterless yellow, black, brown, and blue	TT-P-645	MIL-E-24635 color as specified	1	Exterior of anchor housing; parts of mines
60	Enamel, semigloss blue-gray	TT-P-645	MIL-E-24635 blue, color No. 25109 gray, color No. 26307	1	Metal surfaces of ammunition and ammunition containers
63	Exterior gray enamel	TT-P-645	MIL-E-24635 color No. 26118	3 or more to 6 mils minimum	General usage on aluminum or steel torpedoes
67	Enamel, gloss olive drab	TT-P-645	MIL-E-24635, gloss olive drab, color No. 14087	1	Ammunition items, such as bombs, shells, rockets, and mines; interior and exterior
69	Lacquer, fuel resistant	None	MIL-C-4556, or MIL-P-24441 (epoxy)	1	Internal surfaces of steel fuel tanks
70	Coating, permanent resin	None	MIL-R-3043	1	Baked coating for interior of magnesium fuel tanks, dropped steel tanks, interior of plumbing lines carrying liquids or fire fighting chemicals; parts housed in lubricating hydraulic oil or grease
73	Coating system for areas around alkaline storage batteries	MIL-P-24441	MIL-E-24441, top coat MIL-E-24635	2	Areas around alkaline storage batteries or parts further removed but subjected to spillage or spray
75	Coating, epoxy polyamide, chemical and solvent resistant	MIL-P-53030 (1 coat)	MIL-C-22750 color as specified	1	Interior or exterior surfaces of ordnance equipment subjected to the influence of synthetic lubricants; exterior markings
76	Enamel, gloss color as specified	TT-P-645	MIL-E-24635 color as specified	2 0.8 to 1.2 mils	Interior and exterior surfaces of aircraft, aircraft parts, and related equipment; also suitable for ground equipment and parts where good weather resistance is required

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TABLE I. *Naval Sea Systems Command standard paint systems* - Continued.

System No.	Name and Color	Primer, Filler, Surfacer[1]	Paint and Color No. FED-STD-595	No. of Coats	Uses
[2]78	Compound, edge sealing	None	MIL-S-8802	1	Edge sealing of plastic or ceramic insulators for radio antennas and the like, after installation in exterior locations
80	Paint, silicon phosphate or silicon zinc	MIL-P-24441	MIL-P-24441	4 maximum for system	Used over abrasive blasted steel surfaces (down to the white metal) on steel shipboard tanks used for fresh and salt water ballast.
80A	Paint silicate, plus vinyl or epoxy coating	DOD-P-24648, Class 3 system	MIL-P-24441, Coat with a vinyl or epoxy paint (alkaline resistant) as required for color, or MIL-E-24635	1	Guns or gun mounts

[1]MIL-P-53030 Epoxy Primer may be used in lieu of TT-P-645 for galvanic corrosion.
[2]No change was required for environmental problems with this system (see 6.3).

TABLE II. *Index to paint systems in table I.*

Type of equipment, weapon, surface, or exposure	System No.
Acid resistance	47, 73
Ammunition	
Exterior metal surfaces and containers	52, 07
Inert or dummy	07
Anchor housing	
Exterior	58
Antistatic	73
Armored cable	25
Battery racks and areas around acid storage batteries	47, 73
Bombs	07, 67
Bomb bays	14
Exteriors, before loading explosives	67
Boxes	
Accessories, wooden	31
Ammunition	25
Cartridge	25
Hardwood	31
Metal	22

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TABLE II. *Index to paint systems in table I* - Continued.

Type of equipment, weapon, surface, or exposure	System No.
Wooden (exposed on topside)	22
Buoys	49
Cables	37
Camouflage finish, good weather resistance	58
Cases	
Instrument	13, 14
Metal	25
Mine, interior	13
Torpedo test equipment	22
*Cavities in contact with explosives	47
Color coding of weapons and ammunition components	54
Crates	07
Deck, below	37
Depth charges	22, 25
Distance sighting	51
Electrical panels and enclosures	37
Electrical parts	
Moisture-proofing	42
Sealing	42
Touch-up	42
Exterior markings	75
Fire control equipment	
AN/SPQ-9	20
AN/SPQ-9B	20
AN/SPG-60	20
Below decks	37
Enclosed	37
Reflective surfaces	20A
Radomes	20B
Flight gear	07
Floats	48, 49
Drill, aluminum	58
Fluorescent finish, for training equipment	51
General purpose, exterior and interior	14
Ground equipment and parts for good weather resistance	76
Gun barrels	22
Exterior	25
Harbor defense items	49
High light reflection	14
High temperature exposure, heat ducts, parts, components, and surfaces	25, 76
Instrument racks	07
Instruments	
Portable	37

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TABLE II. *Index to paint systems in table I* - Continued.

Type of equipment, weapon, surface, or exposure	System No.
Launchers	22
Lettering and stenciling	54, 48, 48A
Magnesium fuel tanks, interior	70
Mines	
*Metal not exposed to weather, for high light reflectance fire retarding	14, 14C
Anchors	07, 48
Exterior cases before loading with explosives	07, 67
Floats	48
Moored	48
Stenciling and marking of moored mines	48, 48A
Missiles	
Exterior, where specified	13, 58
Rain-erosion resistance	73
Synthetic oil resistance	14, 58
Warheads	14
Mounts, gun	22, 80A
Enclosures, interior surfaces	14, 25
Name plates	07
Optical instruments	
Exterior surfaces	07
Exterior transporting cases	20, 37
Interior surfaces	07
Knobs, eyepieces	07
Parts housed in lubricating hydraulic oil or grease	70
Plumbing carrying liquids or fire fighting chemicals	70
Projectiles	07, 67
Projectors on shipboard topside	22
Propellant containers	07
Radio antennas, plastic or ceramic insulators, edge sealing	78
Rain erosion surfaces	73
Rocket heads, where specified	07
Before loading with explosives	67
Rockets	67
Shells, before loading with explosives	67
Shields, before loading with explosives	14, 25
Equipment in shields	37
Surfaces, exterior	
Cavities in contact with explosives	47
Metal not exposed to weather, for high reflectance	14, 14C
Fire retarding	14C
Shipboard, topside	22, 25
Steel tanks	69
Droppable	70

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TABLE II. *Index to paint systems in table I* - Continued.

Type of equipment, weapon, surface, or exposure	System No.
Internal surfaces of fuel tanks	69
Synthetic lubricant exposure, interior or exterior surfaces	70, 75
Tanks	25, 69
Torpedoes	53
*Aluminum exterior surfaces	52, 63
For training	51
*Exercise heads	51
For visibility	51
Interiors, where specified	14
Steel parts	58
Training ordnance equipment, fluorescent	51
Turrets	37
Underwater or floating objects	51

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TABLE III. *Discontinued, altered, or replaced paint systems.*

Former system No.	Current System No.	Changes
All except 47 & 78		Low VOC specifications substituted where necessary
09	—	Discontinued; use system 07
10	—	Discontinued; use system 25
12	—	Discontinued; use system 22
14A	—	Discontinued; use system 14
14B	—	Discontinued; use system 14
17	—	Discontinued; use system 07
27	—	Discontinued; use system 14
33	—	Discontinued; use system 31
36	—	Discontinued; use system 31
37A	—	Discontinued; use system 37
46	—	Discontinued; use system 07
46A	—	Discontinued; use system 07
50	—	Discontinued
50A	—	Discontinued; use system 07
55	—	Discontinued; use system 07
56	—	Discontinued; use system 14
56A	—	Discontinued; use system 58
57	—	Discontinued
59	—	Discontinued; use system 07
62	—	Discontinued; use system 13
64	—	Discontinued; use system 51
65	—	Discontinued; use system 58
66	—	Discontinued; use system 48
68	—	Discontinued
71	—	Discontinued; use system 76
72	—	Discontinued; use system 73
74	—	Discontinued; use system 73
77	—	Discontinued; use system 58
79	—	Discontinued; use system 51
79A	—	Discontinued; use system 51
81	—	Discontinued

5.3 Nonstandard paint systems. Use of paint systems other than those shown in table I shall be subject to prior approval by the Naval Sea Systems Command. Systems employing materials and processes covered by Federal and military specifications and standards will be given preference. Use of new or improved materials or processes will be approved if it is shown that such use affords better protection or greater economy. Results of tests by standard methods or properly controlled experimental procedures will be required. Where energetic materials come into contact with painted surfaces, as a minimum, chemical and physical compatibility data must be obtained. Specification data suitable for

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inclusion in Federal or military specifications shall be furnished as soon as practicable to enable procurement without use of proprietary designations.

5.4 Paint systems for ordnance.

5.4.1 Shipboard topside. Preparation for painting shall be in accordance with 5.1. Fixed ordnance on shipboard topside, which is exposed to view from sea or air, shall be painted in accordance with table II on all exterior surfaces.

5.4.2 Unexposed surfaces. Exterior surfaces of ordnance equipment not normally exposed to view from sea or air shall be painted aluminum, system 25, white, system 14, or light gray, system 37. This applies to both portable and fixed equipment within shields, turrets, or other enclosures, or below deck. Refer to table I for other systems and their uses.

5.4.3 Interiors. Interior surfaces of gun mount enclosures or shields shall be painted gloss white, system 14. The final coats may be applied after installation. Refer to table I for other paint systems and their uses.

5.5 Paint systems for weapons. All weapons and their parts shall be properly cleaned and prepared for painting in accordance with MIL-F-7179. Unpainted surfaces shall be treated in accordance with 4.5. Refer to table II for paint systems suitable for various types of weapons and exposure conditions.

5.5.1 Color-coding. Color-coding used for identification of types or uses of various weapons or their fillers is not covered herein. Reference should be made to current specifications for painting of particular weapons or to color-coding standards. Any color of paint listed for system 54 in table I may be used for color-coding.

5.5.2 Missiles and rockets. Color-coding shall conform to MIL-STD-709. Paint systems approved for use on missiles, rockets, and their various components are listed in table I. Reference should be made to MIL-H-18911 and MIL-P-21960. Warheads shall be painted in accordance with applicable specifications such as MIL-H-18911 and MIL-P-21960, and drawings. Interior surfaces, which are in contact with high explosives, shall be coated in accordance with system 47 or a suitable alternative material that has been approved in accordance with 5.3.

5.5.3 Bombs. Bombs shall be painted as specified in the applicable drawings. System 07 or 67 may be used. Paints for color-coding are listed under system 54.

5.5.4 Torpedoes. For painting of torpedoes and their components, reference may be made to systems 14, 51, 52, 53, or 63, as applicable.

5.5.5 Mines and depth charges. Painting of mines depends on the types of mines (moored or bottom). Systems 48, 49, and 67 are used where the greatest anti-fouling and protection against water are required. System 07 is used where less severe conditions are expected. System 47 is used for explosive cavity lining and system 54 is used for color-coding. Depth charges and projectors exposed on shipboard topside shall be painted haze-gray, system 22, or aluminum, system 25.

5.5.6 Gun ammunition. Until color codes are standardized, gun ammunition shall be painted in

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accordance with current specifications, such as MIL-P-18948 or other applicable documents.

5.6 Sealing and painting procedures. Various details in connection with sealing and painting procedures, such as sealing against water entry, drainage, protection of metals against electrolytic corrosion, conditions for painting, and details of paint application are described below.

5.6.1 Drainage and water seals. Considerable damage to equipment may result from retention of water, particularly salt water, in cavities, cracks, recesses, depressions, or on level surfaces. It is very important to guard against damage by proper design and by use of sealing materials where necessary. Drainage holes shall be provided wherever horizontal surfaces might retain water. Irregularities that prevent free drainage may be smoothed by applying as many coats of surfacer conforming to TT-P-662 as necessary over the primer coat, and sanding until smooth. Rough or pitted surfaces may be smoothed by filling with cement conforming to DOD-C-24176 after phosphating or pretreatment. Seams, cracks, or openings at joints should be filled with caulking compound conforming to MIL-C-18255 after priming.

5.6.2 Prevention of electrolytic corrosion. To minimize corrosion due to electrolytic action between dissimilar metals, as defined in MIL-STD-889, such metals should not be placed in direct contact with each other. When it is necessary to place dissimilar metals close together, they shall be insulated from each other as indicated in 5.6.2.1.1, and exposed edges sealed to prevent contact with salt water which greatly accelerates electrolytic action.

5.6.2.1 Treatment of faying surfaces. Surfaces in close contact are termed faying surfaces. When faying surfaces move with respect to one another or are enclosed within brackets or cases containing lubricants or hydraulic fluids, no additional protection is required. Grease conforming to MIL-G-81322 may be applied for protection during shipment or storage, except when grease conforming to MIL-G-23827 is specified for the particular equipment.

5.6.2.1.1 Metal surfaces with noncritical dimensions. Unless otherwise specified (see 4.5.1) or where dimensions or fits are critical (as in optical instruments), faying surfaces of both similar and dissimilar metals to be permanently assembled shall be painted on each surface with two coats of primer conforming to TT-P-645. Allow each coat to dry hard before the next coat is applied. When metals are dissimilar as defined in MIL-STD-889, insulation tape shall be applied a minimum of 20 mils thick, in addition to primer coats over the entire area of each surface. The insulation tape shall be 0.010 inch thick and shall conform to MIL-I-24391 or other approved equivalent specification such as MIL-T-23142. After surfaces are joined, any openings shall be sealed with caulking compound conforming to MIL-C-18255 and well covered with the specified coat of paint to prevent entrance of water.

5.6.2.1.2 Metal surfaces of critical dimensions. Faying surfaces of parts having critical dimensions, such as optical and fire-control instruments, are normally protected against moisture and should not be painted. Parts that are to be in contact with instrument grease conforming to MIL-G-23827 may be coated before assembly. Optical parts shall be assembled only with specified cements or sealers.

5.6.2.1.3 Wood. Two coats of varnish conforming to TT-V-119 shall be applied to surfaces of wood that will be in contact with metal. Two coats of primer conforming to TT-P-645 shall be applied to the metal surfaces. If required by the exposure conditions, wood may be treated with water-repellant preservative conforming to Type II, Composition A, of TT-W-572 prior to painting.

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5.6.3 Conditions for painting.

5.6.3.1 Atmospheric conditions. Painting should be performed when the air temperature is between 60 and 90°F and the relative humidity is not over 65 percent. Dusty or windy conditions should be avoided to prevent contamination of freshly painted surfaces.

5.6.3.2 Preparation of surfaces. Thorough surface cleaning and preparation in accordance with MIL-F-7179 are prerequisites for acceptable painting. Surfaces shall be thoroughly dry. It is preferred that any welding, brazing, or soldering operations be performed before painting. However, if any of these operations are performed after painting, the surfaces within the heat-affected area shall be sanded, cleaned, reprimed, and repainted in accordance with this standard. When painted surfaces have been marred, abraded, or show rust, the affected areas shall be recleaned, reprimed, and repainted in the same manner as the original surfaces.

5.6.4 Application of coatings. Unless otherwise specified, each pretreatment, primer, and paint coat shall be allowed to dry hard before the next coat is applied. Paint coverage shall be uniform and unbroken. Methods of paint application (brushing, spraying, or dipping) shall be in accordance with good workmanship. Thinning, baking, drying, mixing, or other procedures shall be in accordance with applicable specifications for the materials used or for the items to be painted.

5.6.5 Paint thickness. For naval ordnance, it is desirable to prevent buildup of heavy paint coats. However, certain minimum thicknesses are recognized as desirable for optimum protection under specified conditions, exposures, or deteriorating influences. The thickness of each coat or paint system shall be in accordance with the applicable specification or drawing. In general, application and control shall be in accordance with MIL-F-18264.

5.6.6 Colors. Color numbers shall be in accordance with FED-STD-595. Colors of final paint coats shall be in accordance with the applicable specifications for the paint, enamel, or varnish, the applicable specifications and drawings for the item, MIL-STD-709, or this standard. Colors of primers not specified herein shall be in accordance with the applicable specification for the item.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use.

6.2 Issue of DODISS. When this standard is used in acquisition, the applicable issue of the DODISS must be cited in the solicitation (see 2.1.1).

6.3 Environmental damage problems. Specifications for paints which would violate EPA regulations were deleted from table I and replaced with specifications with low volatile organic compounds where necessary.

6.4 Suggested source. Spraylat 599-Y1002 has been obtained from:

Spraylat Corporation
716 South Columbus Avenue
Mt, Vernon, NY 10550

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.6 Subject term (keyword) listing.

Bombs
Depth charges
Gun ammunition
Mines
Missiles
Rockets
Volatile organic compounds

Preparing activity:
Navy-OS
(Project MISC-N174)

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-STD-001303C(OS)	940411

3. DOCUMENT TITLE

PAINTING OF NAVAL ORDNANCE EQUIPMENT

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
COMMANDER NAVAL SURFACE WARFARE CENTER (CODE 8420)	(1) Commercial 301-743-4510/4253	(2) AUTOVON DSN 354-4510/4253
c. ADDRESS (Include Zip Code) 101 STRAUSS AVENUE INDIAN HEAD, MD 20640-5035	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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